$$\frac{v_{out}}{v_{in}} = -\beta R_C \left[\frac{R_B (R_B \| R_S)}{R_S + R_B} - \frac{1}{R_E} \right] v_{in}$$

$$= -100 \times 3 \left[\frac{830(\quad)}{831} - \frac{1}{1} \right]$$

$$= -300 \left[\left(\frac{830}{831} \right)^2 - 1 \right]$$

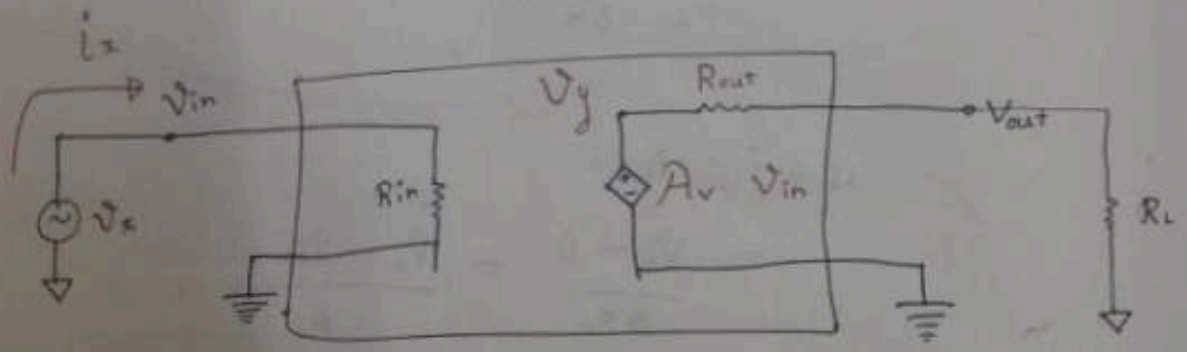
$$= -300 [\quad]$$

$$R_B \| R_S = \frac{830}{831} \qquad = -0.721$$

[Small Signal]

Equivalent Model of Amplifier:

R_{out}, R_{in}, A_v



R_{in} → Input Impedance

R_{out} → Output Impedance

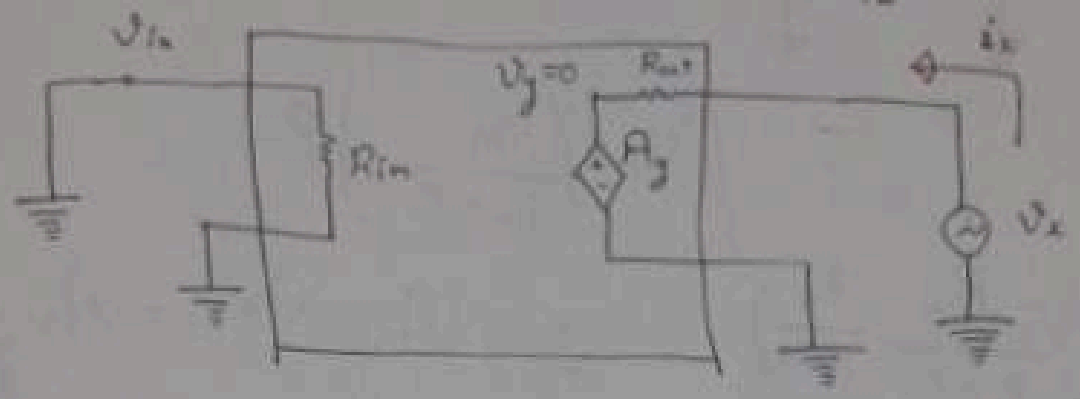
A_v → Gain

Apply v_x at input & find out i_x,

$$R_{in} = \frac{v_x}{i_x}, \quad R_L \to \infty$$

Apply v_x at input & find v_{out} with $R_L \to \infty$

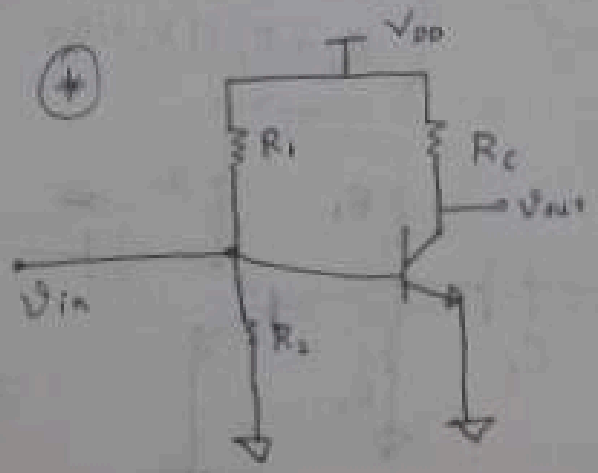
$$A_v = \left.\frac{v_{out}}{v_x}\right|_{R_L \to \infty}$$



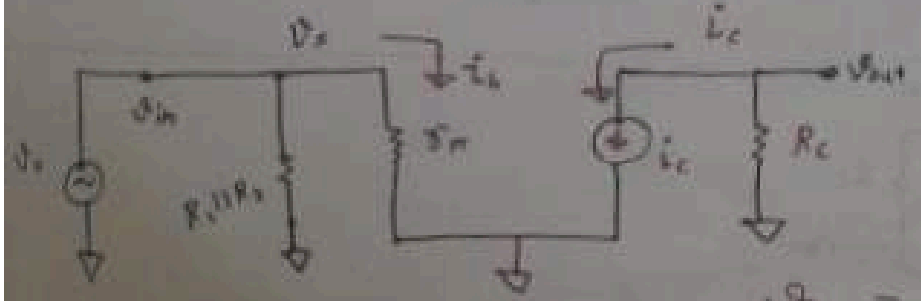
Apply v_x at output & find i_x with $v_{in} = 0$,

$$R_{out} = \left.\frac{v_x}{i_x}\right|_{v_{in}=0}$$

#



(not physically grounded, grounded using a capacitor)



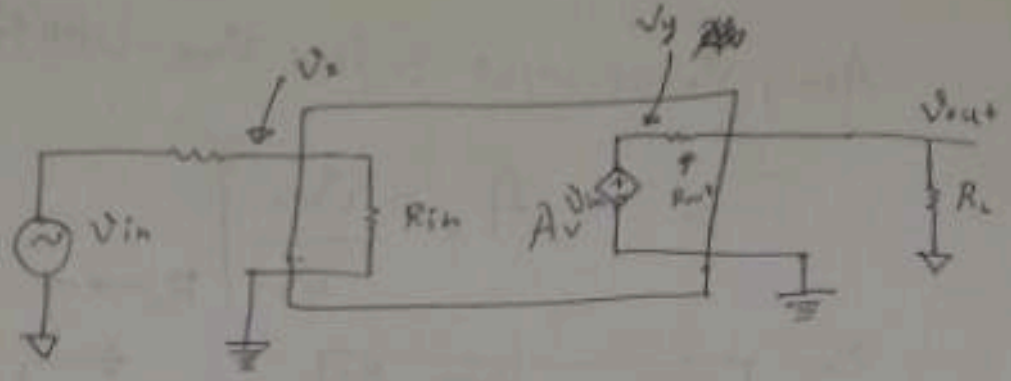
$$i_b = \frac{v_x}{r_\pi}, \quad i_c = \beta \frac{v_x}{r_\pi}$$

$$v_{out} = -i_c R_C$$

$R_{in} = R_1 \| R_2 \| r_\pi$

$R_{out} = R_C$

$$\left[A = \frac{v_{out}}{v_{in}} = \frac{-\beta R_C}{r_\pi}\right] \quad v_{out} = -\frac{\beta v_x}{r_\pi} R_C$$



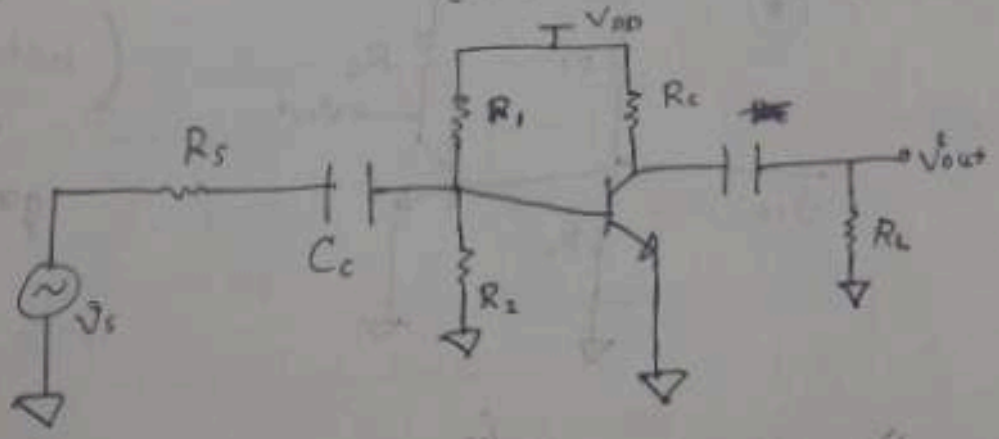
$$\frac{v_{out}}{v_{in}} = A_v \cdot \frac{R_{in}}{R_s + R_{in}} \cdot \frac{R_L}{R_L + R_{out}}$$

$$v_x = \left(\frac{R_{in}}{R_{in} + R_s}\right) v_{in}$$

$$v_y = A_v \, v_x = A_v \frac{R_{in}}{R_s + R_{in}} v_{in}$$

$$v_{out} = \frac{R_L \, v_y}{R_L + R_{out}}$$

$$(A_v)_{Loaded} = \left(\frac{v_{out}}{v_{in}}\right) = \left(\frac{-\beta R_C}{r_\pi}\right) \times \left(\frac{R_1 \,||\, R_2 \,||\, r_\pi}{R_s + R_1 \,||\, R_2 \,||\, r_\pi}\right) \times \frac{R_L}{R_L + R_C}$$



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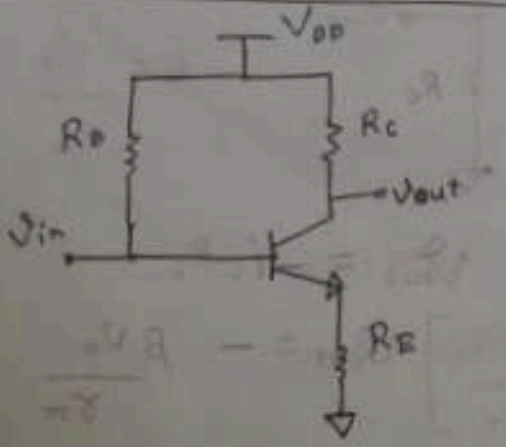
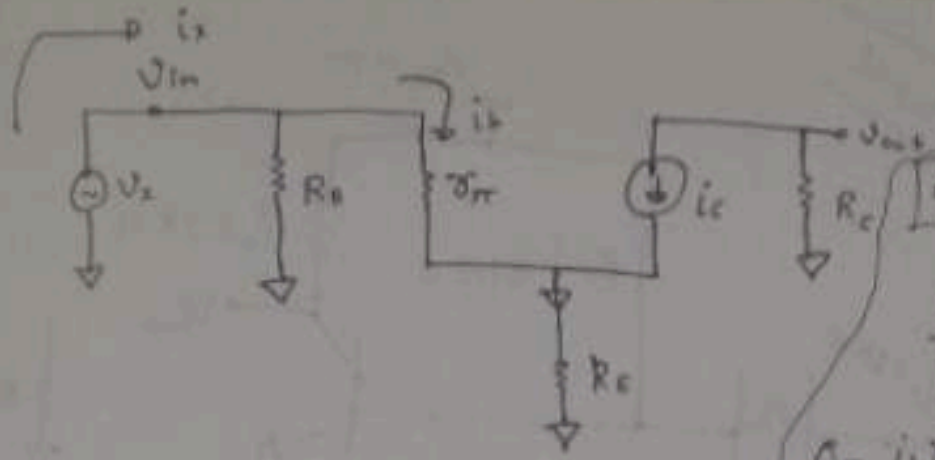




$\cancel{i_c = v_{out}}$

$\boxed{i_{R_c} + i_c = i_x}$

$$\frac{v_x}{R_c} + i_c = i_x$$

$$0 - i_b r_\pi + - i_c R_E = 0$$

$R_{in} \Rightarrow \quad v_b - i_b r_\pi - i_c R_E = 0$

$$v_b = -i_b r_\pi + (1+\beta) R_E \, i_b$$

$$= \left(r_\pi + (1+\beta) R_E\right) i_b$$

$v_b = v_x$

$$\cancel{i_b = \left(\frac{R_B}{R_B + r_\pi}\right) v_{in}}$$

$$\Rightarrow i_b = \frac{v_x}{r_\pi + (1+\beta) R_E}$$

$$i_x = i_1 + i_b = \frac{v_x}{R_B} + \frac{v_x}{r_\pi + (1+\beta) R_E}$$

$$R_{in} = \frac{v_x}{i_x} = R_B \parallel \left(r_\pi + (1+\beta) R_E\right)$$

$$* \; A_v = \cancel{\frac{v_{out}}{v_{in}}} \cancel{\div} \cancel{\frac{v_x}{v_x}} = \frac{v_{out}}{v_x}$$

$$v_{out} = -i_c R_c = -\beta R_c \left(\frac{R_B}{R_B + r_\pi}\right) v_{in}$$

$$\cancel{= -\frac{\beta v_b R_c}{r_\pi}}$$

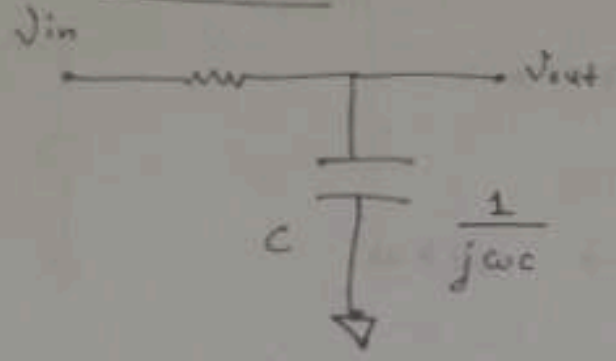
$$= -\beta R_c \times \frac{v_x}{r_\pi + (1+\beta) R_E}$$

$$\frac{v_{out}}{v_x} = -\beta R_c \left(\frac{R_B}{R_B + r_\pi}\right)$$

$$A_v = \frac{-\beta R_c}{r_\pi + (1+\beta) R_E}$$

Band Width

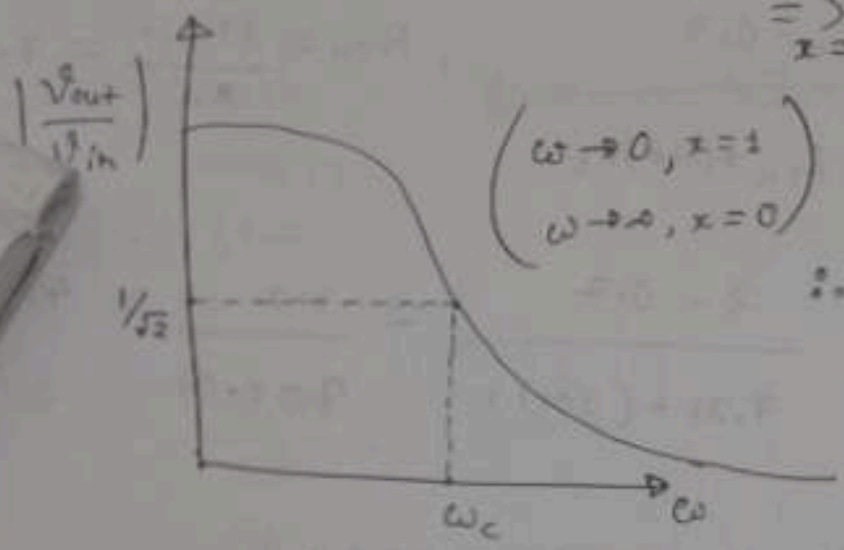
Low-pass filter:



$$v_{out} = \frac{1/j\omega C}{R + 1/j\omega C} v_{in}$$

$$\therefore \frac{v_{out}}{v_{in}} = \frac{1}{1 + j\omega RC}$$

$$\Rightarrow x = \left| \frac{v_{out}}{v_{in}} \right| = \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}}$$



$$\begin{pmatrix} \omega \to 0, & x = 1 \\ \omega \to \infty, & x = 0 \end{pmatrix}$$

$\therefore$ ω_c is the point where my output power is half of input power

~~Band~~

Bandwidth $= f_c - 0$

$= f_c = \frac{1}{2\pi RC}$

$$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}}$$

$$1 + \omega^2 R^2 C^2 = 2 \Rightarrow \omega^2 = \frac{1}{R^2 C^2}$$

(cutoff frequency)

$$\boxed{\begin{aligned} \omega_c &= \frac{1}{RC} \\ f_c &= \frac{1}{2\pi RC} \end{aligned}}$$

High-pass filter:

$$V_{out} = \frac{R}{R + 1/j\omega C} V_{in}$$

$$\frac{V_{out}}{V_{in}} = \frac{j\omega RC}{1 + j\omega RC}$$

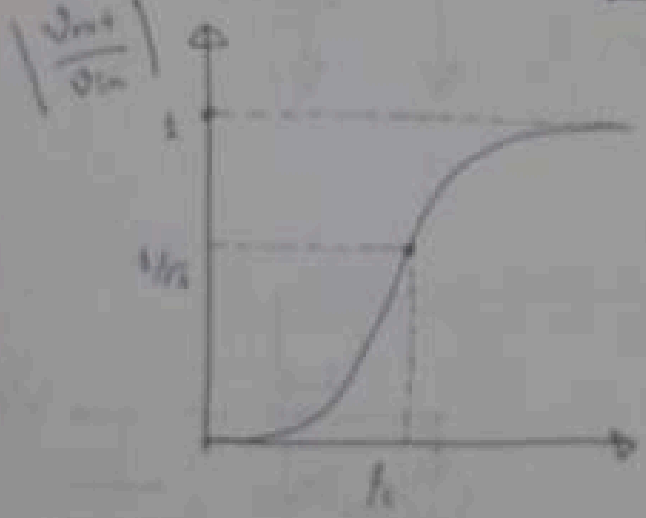
$$\frac{1}{\sqrt{2}} = \frac{\omega RC}{\sqrt{1+\omega^2R^2C^2}} \Leftarrow \left|\frac{V_{out}}{V_{in}}\right| = \frac{\omega RC}{\sqrt{1+\omega^2R^2C^2}}$$

$$\Downarrow$$

$$\left(\begin{array}{l} \omega \to \infty,\ x = 1 \\ \omega \to 0,\ x = 0 \end{array}\right)$$

$$1 + \omega^2R^2C^2 = 2\omega^2R^2C^2 \Rightarrow \omega^2R^2C^2 = 1$$

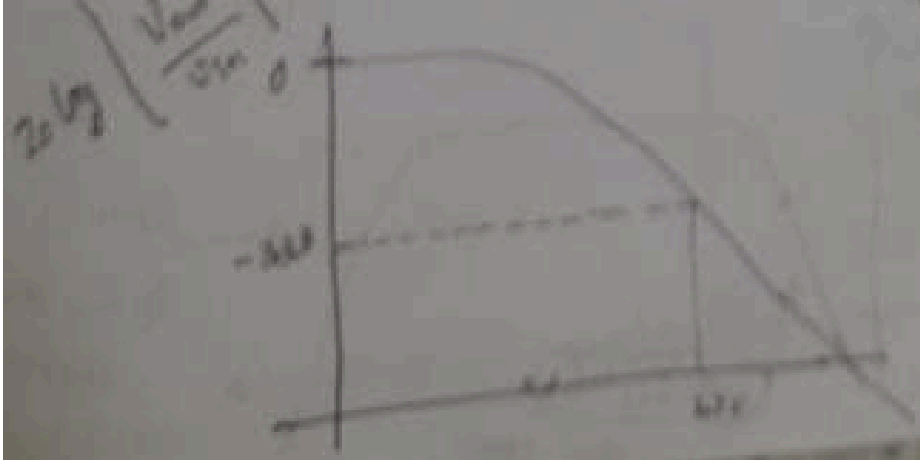
$$\boxed{\omega_c = \frac{1}{RC}} \quad \left(f_c = \frac{1}{2\pi RC}\right)$$



Bandwidth $= \infty - f_c$

(NR) $= \infty$

dB scale $\rightarrow$ $20 \log \left|\frac{V_{out}}{V_{in}}\right|$



$$20 \log \frac{1}{\sqrt{2}} = -3dB$$

High-pass filter:

$$v_{out} = \frac{R}{R + 1/j\omega C} v_{in}$$

$$\frac{v_{out}}{v_{in}} = \frac{j\omega RC}{1 + j\omega RC}$$

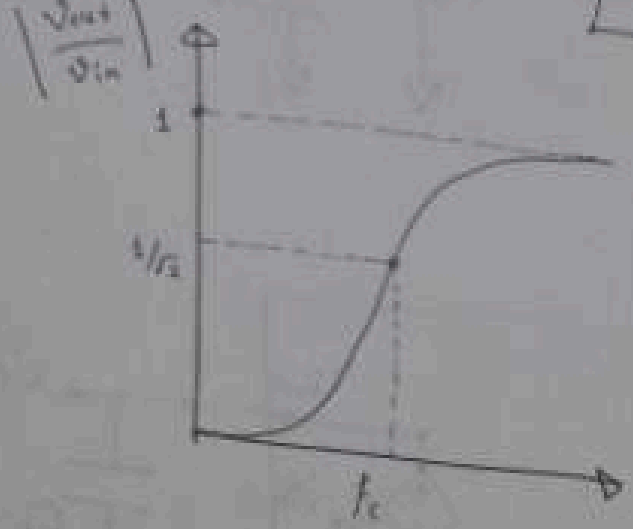
$$\left|\frac{v_{out}}{v_{in}}\right| = \frac{\omega RC}{\sqrt{1+\omega^2R^2C^2}} \Leftarrow \frac{1}{\sqrt{2}} = \frac{\omega RC}{\sqrt{1+\omega^2R^2C^2}}$$

$$\Downarrow$$

$$\left(\begin{array}{l} \omega \to \infty,\ x = 1 \\ \omega \to 0,\ x = 0 \end{array}\right)$$

$$1 + \omega^2R^2C^2 = 2\omega^2R^2C^2 \Rightarrow \omega^2R^2C^2 = 1$$

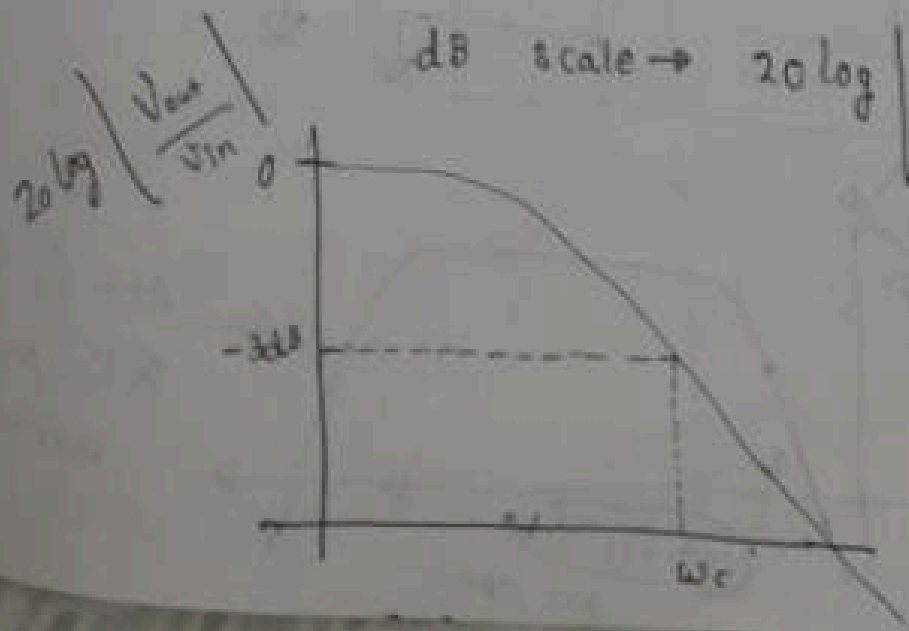
$$\boxed{\omega_c = \frac{1}{RC}} \quad \left(f_c = \frac{1}{2\pi RC}\right)$$



Bandwidth $= \infty - f_c$

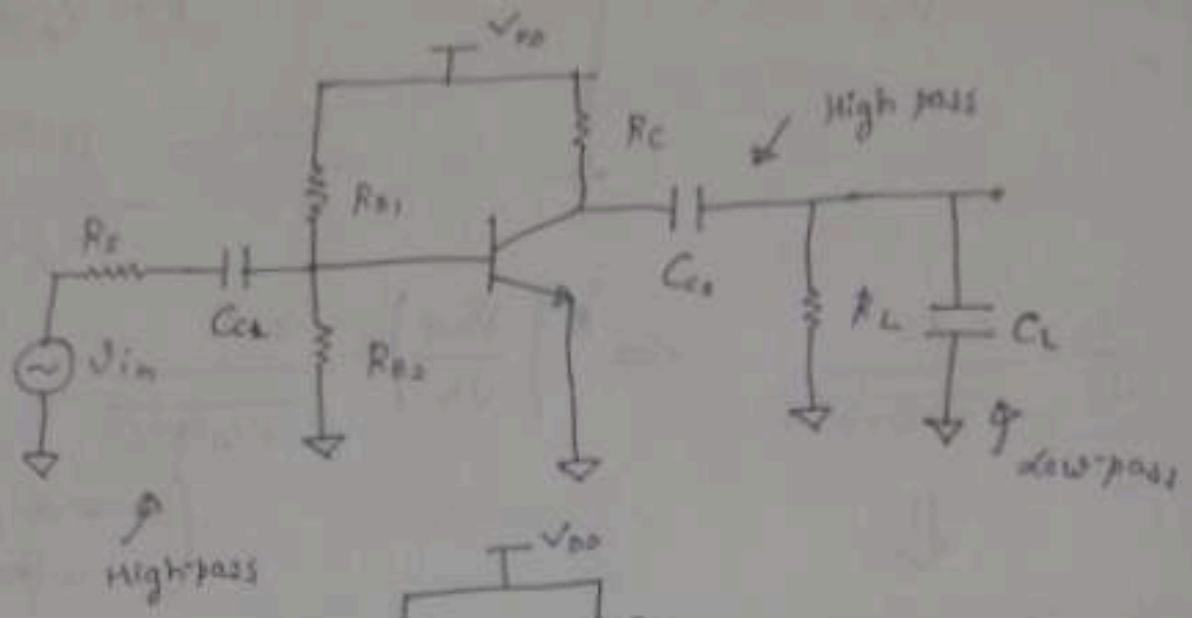
(NA) $= \infty$

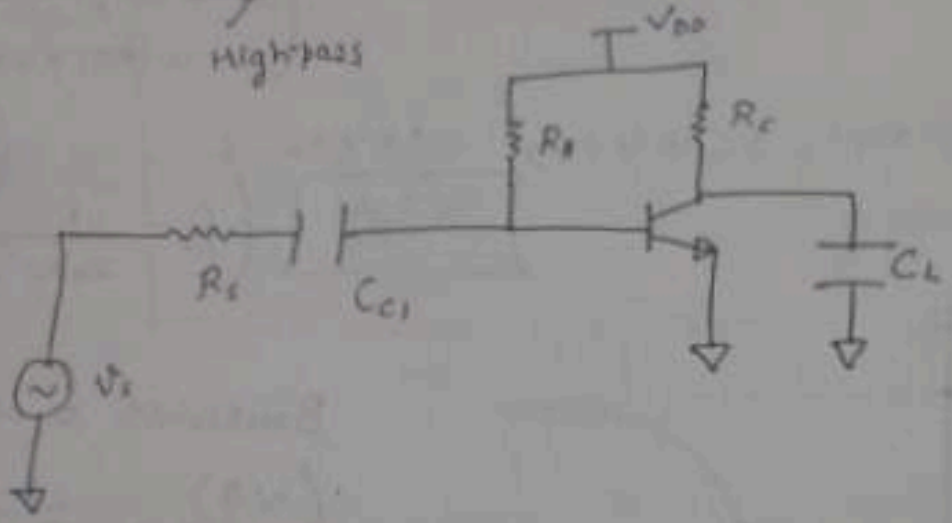
dB scale $\rightarrow$ $20 \log \left|\frac{v_{out}}{v_{in}}\right|$

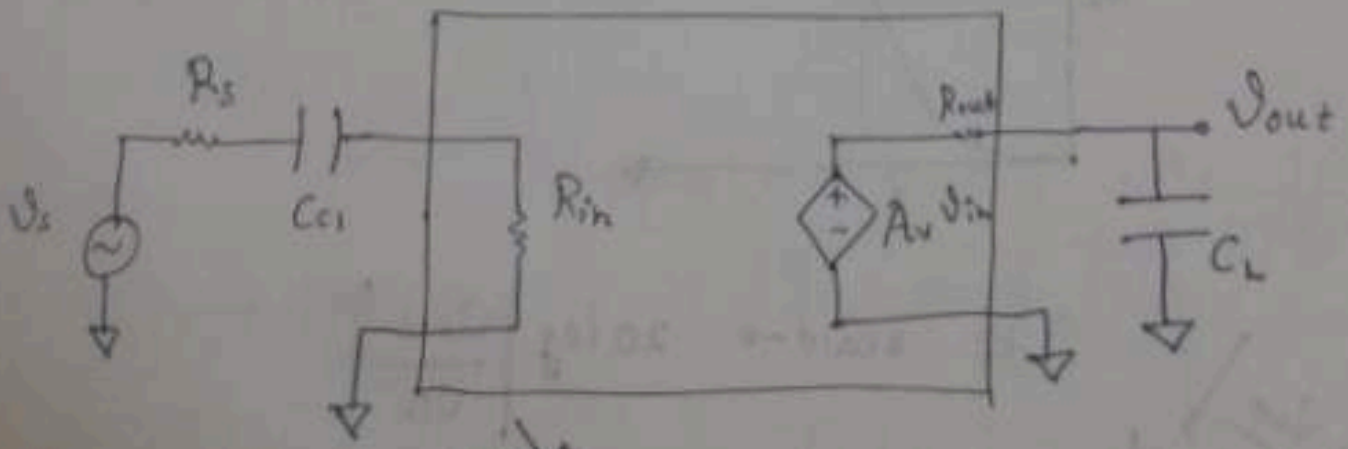


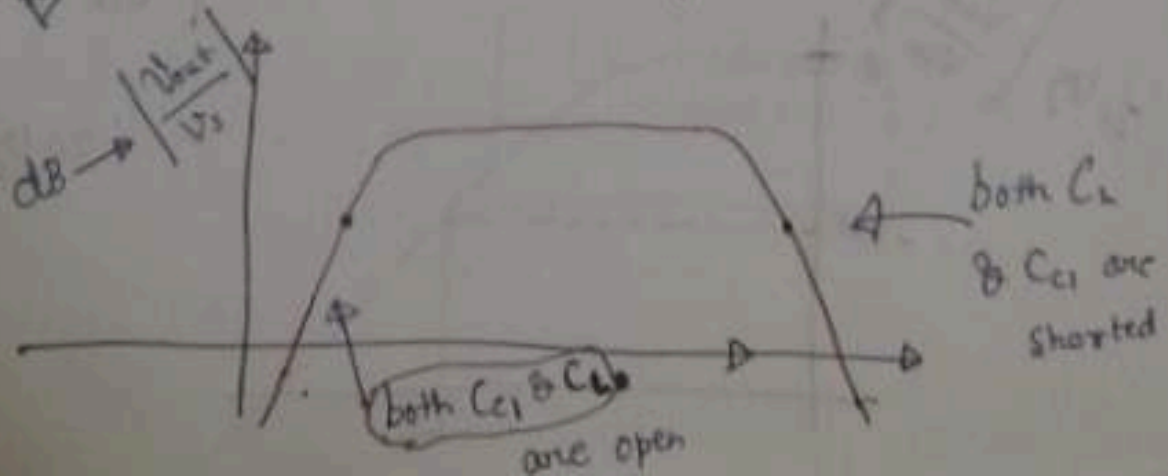
$$20 \log \frac{1}{\sqrt{2}} = -3\,dB$$

Bandwidth of a CEA:





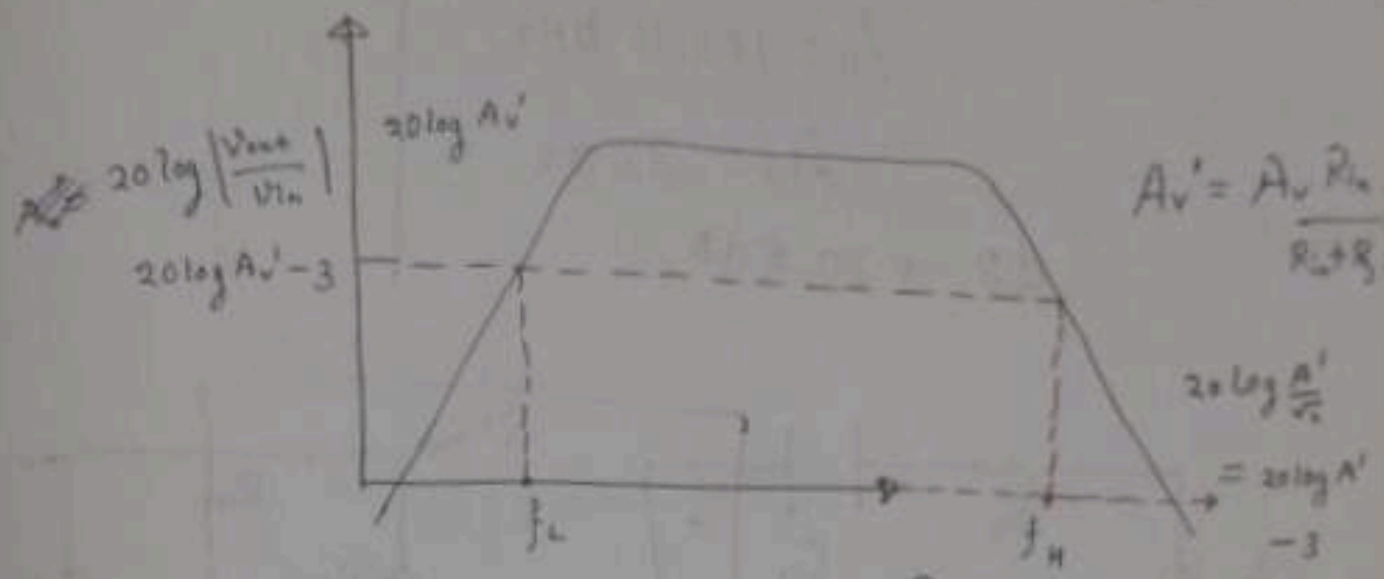




$C_{c_1} >> C_L$

μF $\quad$ nF

midvalue $\Leftarrow$ $A_v' = \frac{v_{out}}{v_{in}} = A_v \frac{R_{in}}{R_{in}+R_s}$

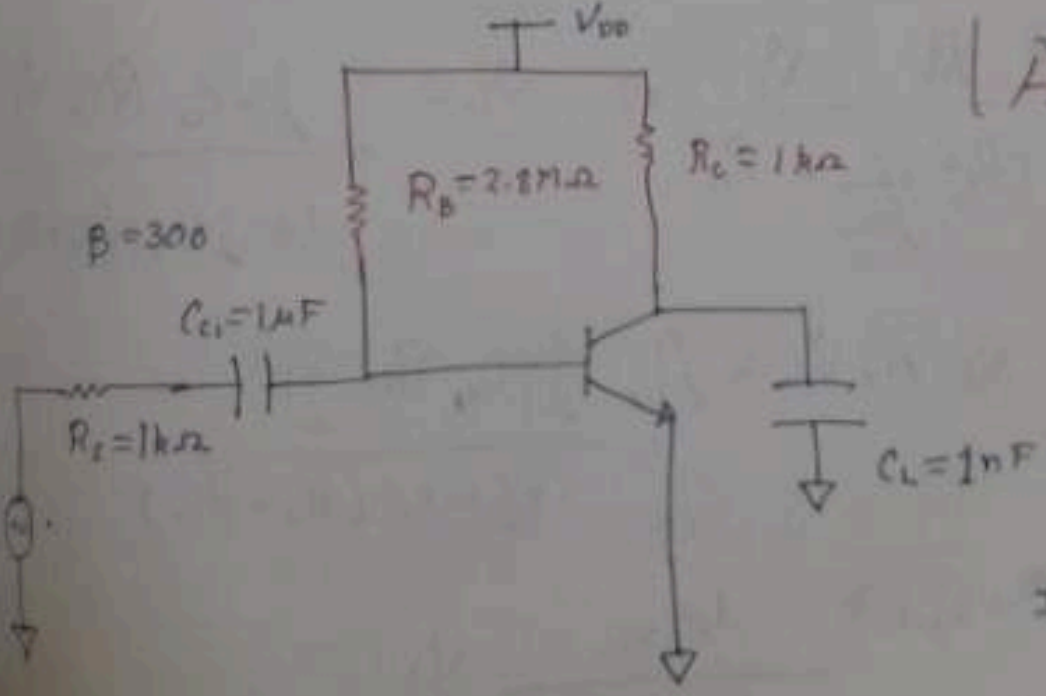


$A_v' = A_v \frac{R_{in}}{R_{in}+R_s}$

$A_v' = A_v \frac{R_{in}}{R_{in}+R_s} \Rightarrow$

$f_L = \frac{1}{2\pi(R_{in}+R_s)C_{c_1}}$, $f_H = \frac{1}{2\pi R_{out} C_L}$

#



$|A_v| = \frac{I_C R_C}{V_T}$

$R_{in} = r_\pi \| R_B$

$R_{out} = R_C$

$I_C = \beta I_B$

$I_B = \frac{V_{DD} - 0.7}{R_B}$

$= 3.32\,\mu A$

dB $\rightarrow 31.65$ $\quad$ $A_v = 38.26$ $\quad$ $I_C = 0.93\,mA$

$r_\pi = 7.83\,k\Omega$

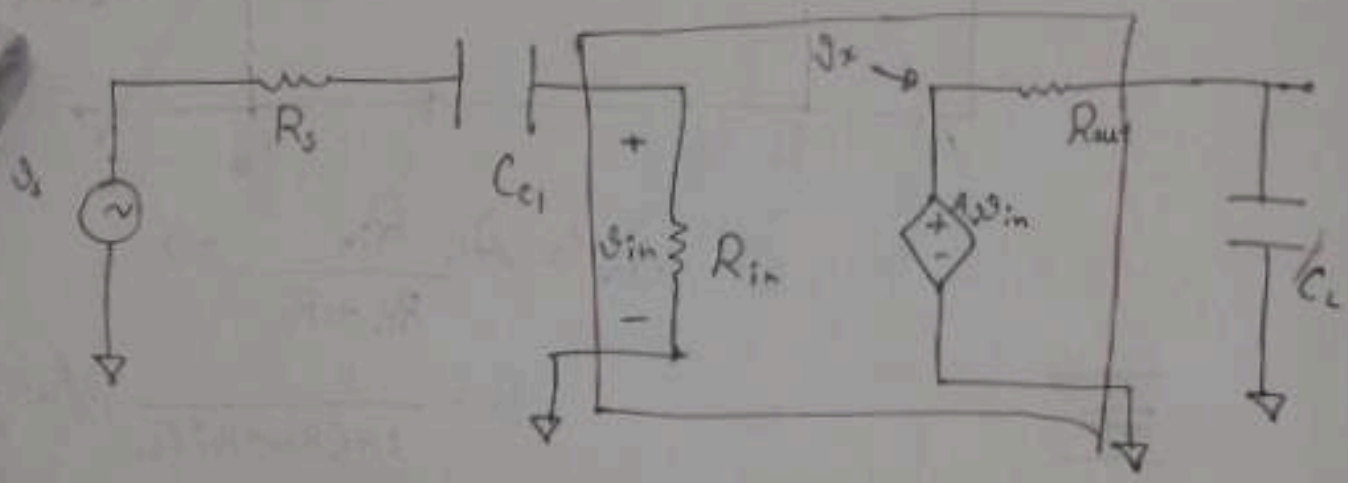
$R_{in} \approx 7.83\,k\Omega$

$f_L = 18.02\,Hz$

$f_H = 159.15\,kHz$

$A_v' = 33.92$

$dB \rightarrow 30.6\,dB$



$$v_{in} = \frac{R_{in} \quad v_s}{R_s + \frac{1}{j\omega C_1} + R_{in}} \Rightarrow v_{in} = \frac{j\omega C_{c_1} R_{in} v_s}{1 + j\omega C_{c_1}(R_s + R_{in})}$$

$$\Rightarrow v_x = A_v v_{in} = A_v \frac{j\omega C_{c_1} R_{in} v_s}{1 + j\omega C_{c_1}(R_s + R_{in})}$$

$$\Rightarrow v_{out} = \frac{1/j\omega C_L}{R_{out} + 1/j\omega C_L} v_x$$

$$v_{out} = A_v \frac{1}{1+j\omega C_L R_{out}} \times \frac{j\omega C_{c_1} R_{in} v_s}{1+j\omega C_{c_1}(R_s+R_{in})}$$

$$\therefore \frac{v_{out}}{v_s} = A_v \cdot \frac{1}{1+j\omega C_L R_{out}} \cdot \frac{j\omega C_{c_1} R_{in}}{1+j\omega C_{c_1}(R_s+R_{in})}$$

$$\left|\frac{v_{out}}{v_s}\right| = |A_v| \frac{1}{\sqrt{1+\omega^2 C_L^2 R_{out}^2}} \times \frac{\omega^2 C_{c_1}^2 R_{in}^2}{\sqrt{1+\omega^2 C_{c_1}^2 (R_s+R_{in})^2}}$$

As, $C_{c_1} >> C_L$

initially, when $\omega^2 = 0 \Rightarrow A_v' = 0$,

$C_{c_1} >> C_L$, where $\omega \to 0$, $\left|\frac{v_{out}}{v_s}\right| \to 0$

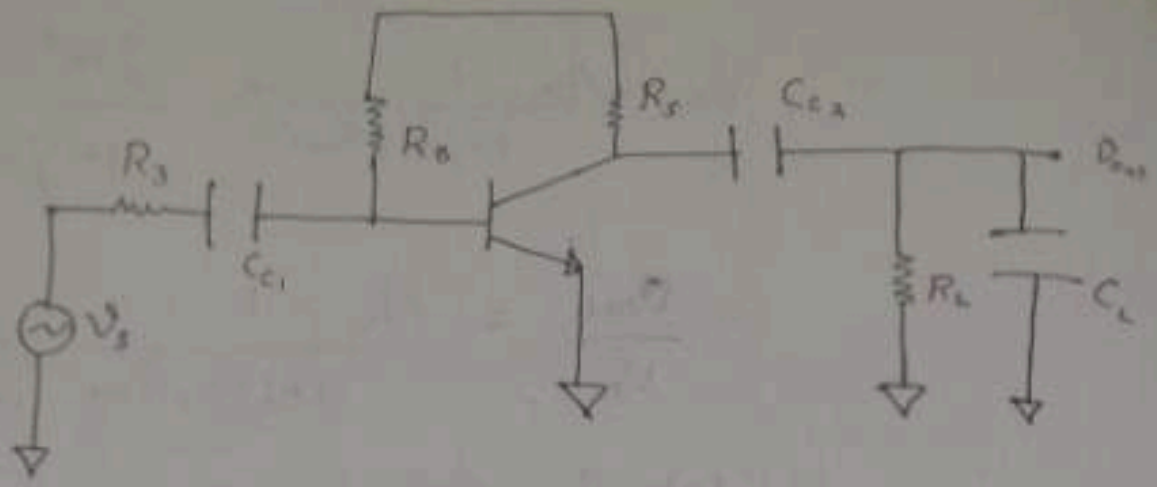
⊛ Mid Range:

(Mid frequency range) $\omega C_L R_{out} << 1$

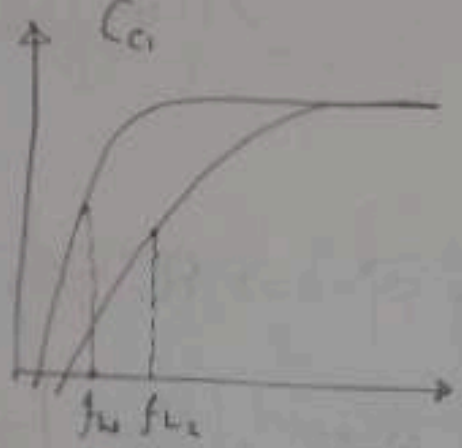
$\omega C_{c_1}(R_{in}+R_s) >> 1$

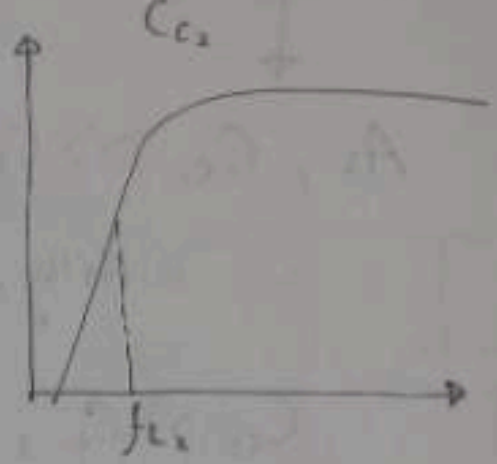
$$\left|\frac{v_{out}}{v_{in}}\right| \approx A_v \cdot \frac{1}{1} \cdot \frac{1 \times \cancel{\omega} R_{in} \cancel{C_{c_1}}}{\cancel{\omega} \cancel{C_{c_1}}(R_s+R_{in})}$$

$$= |A_v| \cdot \frac{R_{in}}{(R_s+R_{in})}$$

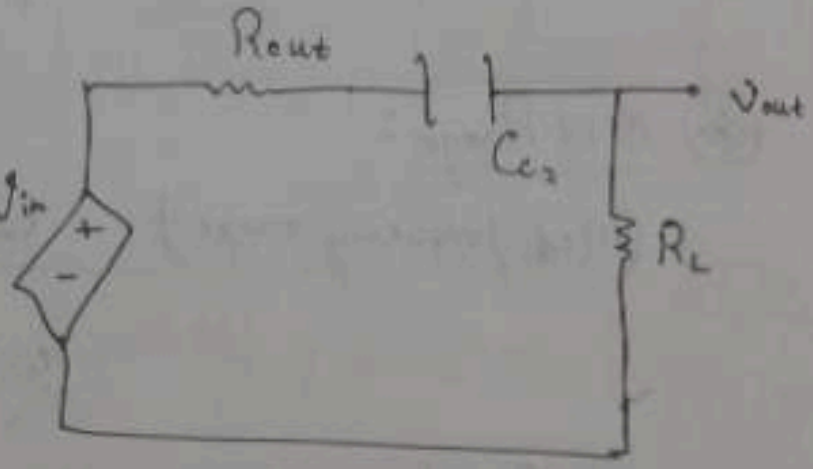


$C_{C_1}, C_{C_2} >> C_L$



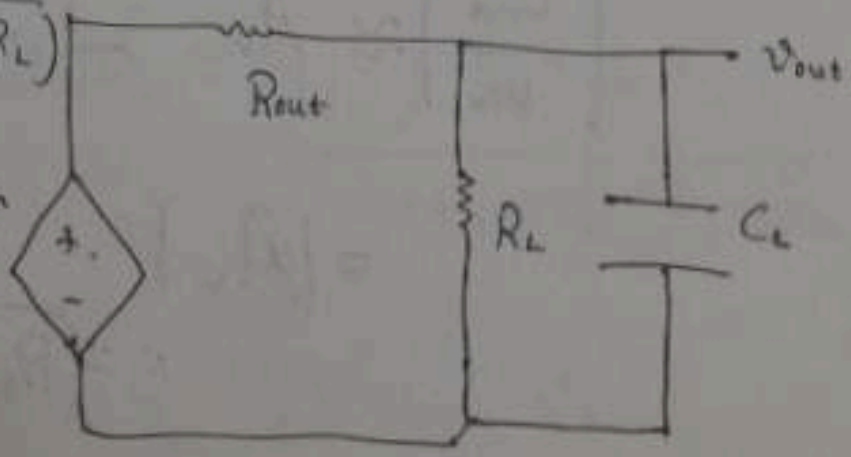


$$f_{L_1} = \frac{1}{2\pi C_{C_1}(R_S + R_{in})}$$

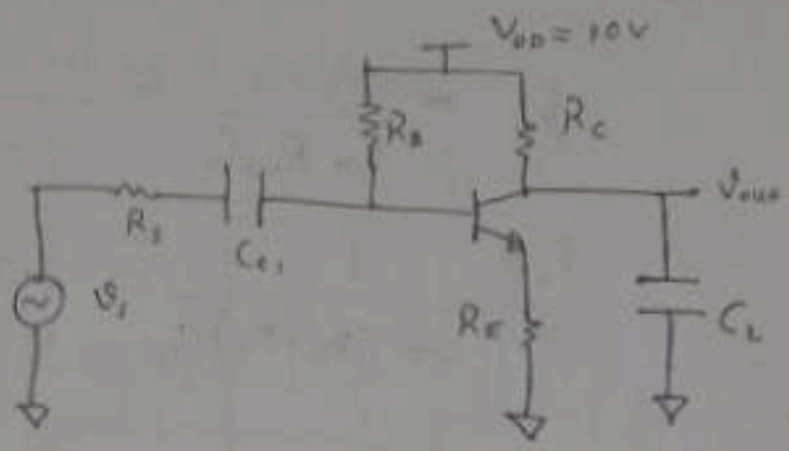


$$f_{L_2} = \frac{1}{2\pi C_{C_2}(R_{out} + R_L)}$$

$$f_H = \frac{1}{2\pi C_L (R_C \parallel R_L)}$$



* Find: the f_L & f_H and midband gain of the following amplifier:



$R_s = 1K\Omega$

$R_C = 4K\Omega$

$\beta = 100$

$R_B = 1.5M\Omega$

$R_E = 1K\Omega$

$C_{c1} = 1\mu F$

$C_L = 10pF$

$$A_v = \frac{-\beta R_C}{r_\pi + (1+\beta)R_E}$$

$$R_{in} = R_B \,//\, \underbrace{(r_\pi + (1+\beta)R_E)}_{105.48k\Omega}$$

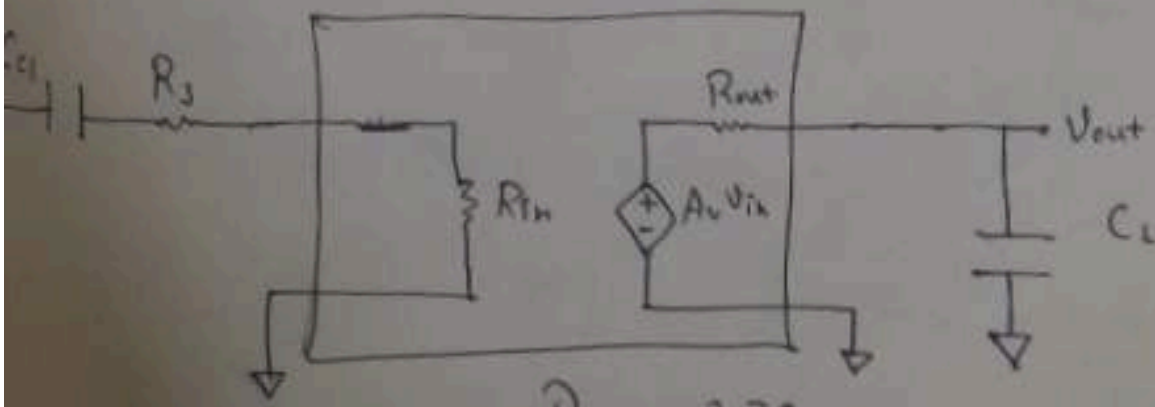
$$R_{out} = R_C = 4k\Omega$$

~~$I_B = 10 - 0.7$~~

$V_{DD} - I_B R_B - 0.7 - I_E R_E = 0$

$$I_B = \frac{9.3}{1.5\times10^6 + (1+100)\times1000} = 5.80\mu A$$

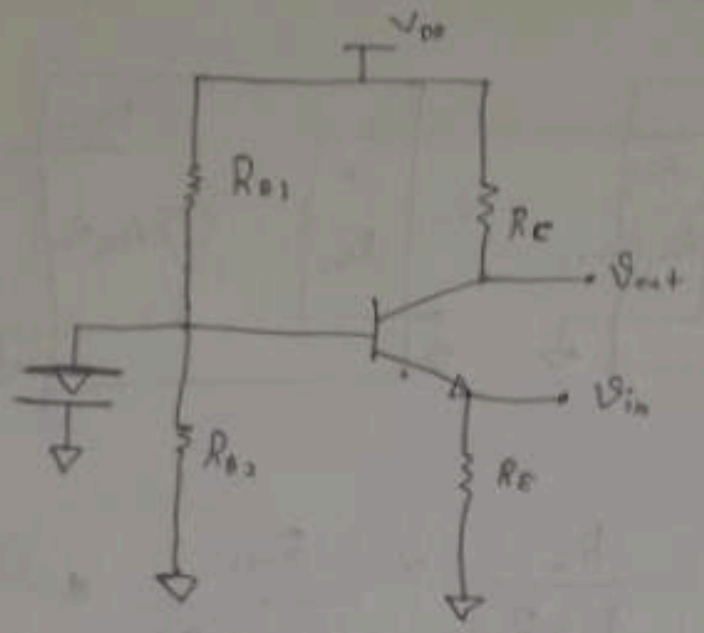
$I_C = \cancel{5.8}$ $0.58mA$

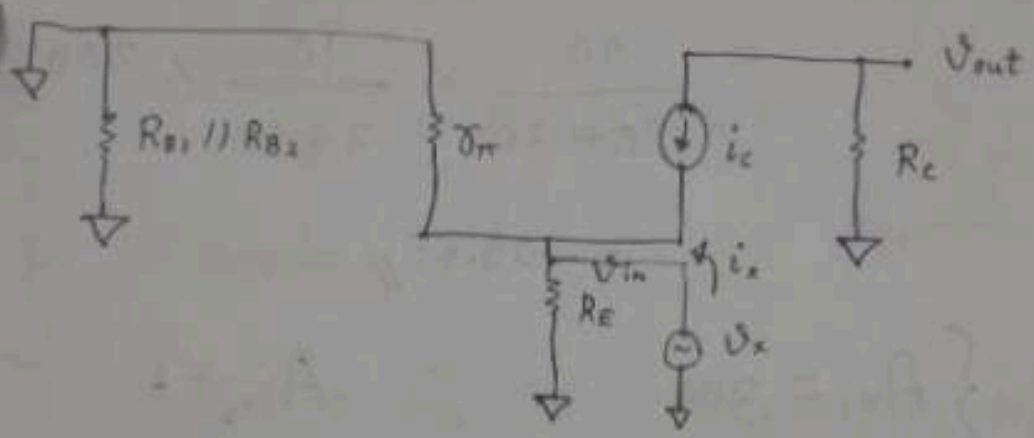


$A_v = -3.79$

$r_\pi = 4.482k\Omega$

$R_{in} = 98.6k\Omega$





$$v_{in} = i_e R_E$$

$$i_e = i_b + i_c + i_x \Rightarrow$$

$$\frac{v_{in}}{R_E} = -\frac{v_{in}}{r_\pi}(1+\beta) + i_x$$

$$v_{out} = -i_c R_c \simeq -\beta i_b R_c$$

$$= -\beta$$

$$\frac{v_{in}}{R_\pi} = i_b + \beta i_b +$$

$$-+ i_b r_\pi - i_e R_E = 0$$

$$i_b r_\pi + i_e R_E = 0$$

$$i_b r_\pi + (1+\beta) i_b R_E + i_x R_E = 0$$

$$i_b (r_\pi + (1+\beta) R_E) +$$

$$v_{out} = - i_c R_c = - \beta i_b R_c$$

$$i_b + i_c + i_x = i_e$$

$$i_e = v_{in} R_E \qquad (1+\beta) i_b + i_x = i_e$$

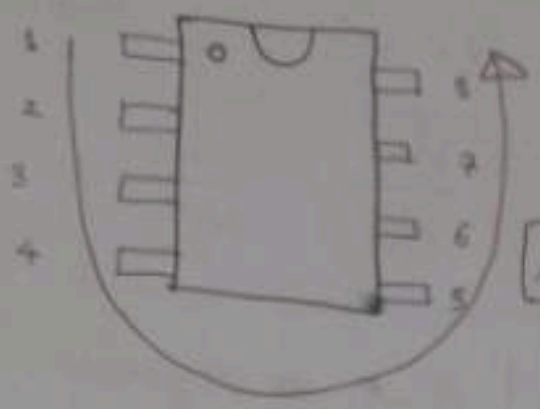
$$R_{out} = R_c , \qquad R_{in} = \frac{R_E r_\pi}{r_\pi + (1+\beta) R_E}$$

$$R_{in} = R_E \,//\, \left(\frac{r_\pi}{1+\beta}\right) , \quad R_{out} = R_c$$

$$i_b = - \frac{v_x}{r_\pi} , \qquad v_{out} = - i_c R_c = - \beta i_b R_c = + \beta \frac{v_x R_c}{r_\pi}$$

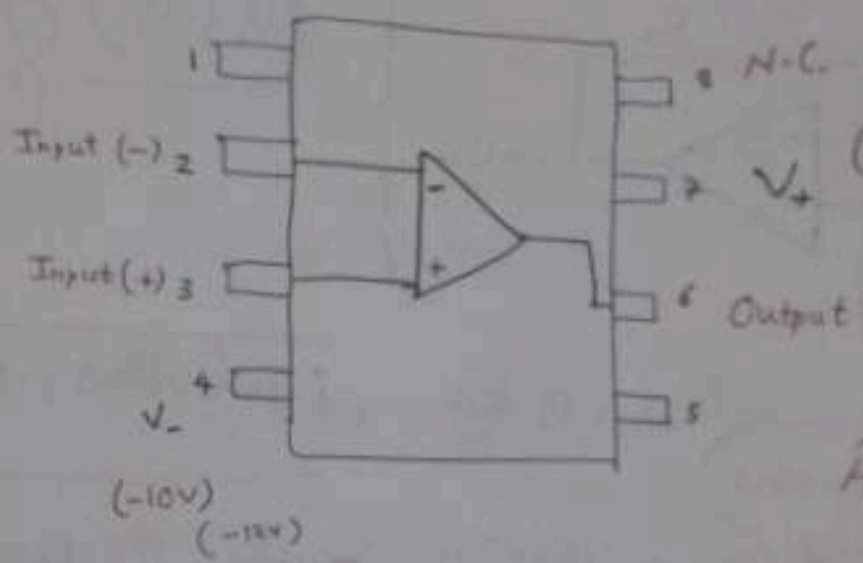
$$\therefore \frac{v_{out}}{v_x} = \beta \frac{R_c}{r_\pi}$$

Op-Amps (Operational Amplifier)



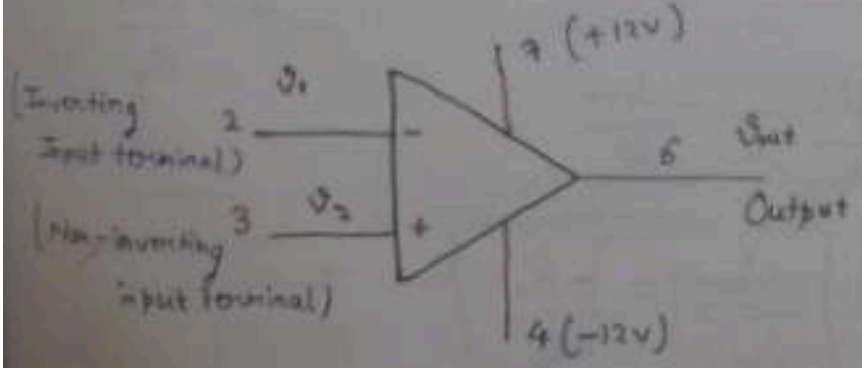
(8-pin IC)
↓
Integrated Circuit

[Any IC numbering]



(20 BJT's)

Additional features
(Off set null)
[1 & 5]



(2 to 5)

$$v_{out} = A_v (v_2 - v_1)$$

(if $v_2 = 0$) → $v_{out} = -A_v v_1$

Similar to CEA

* A_v is very high -10^5
* Amplifies the difference.
* Input Impedance of is very high > 10MΩ
* Output Impedance is very small (10 some Ω's)

$$v_{out} = A_v(\underbrace{v_2 - v_1}_{1mV})$$

$V_{out} = 100V$ ✗

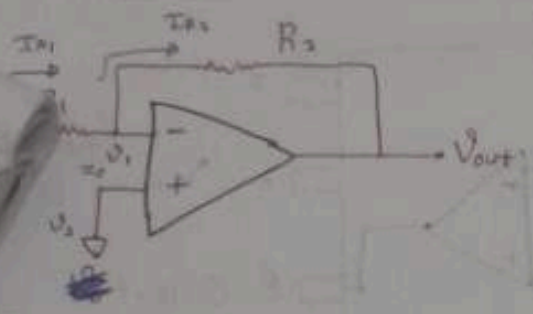
$12V$ ✓

$v_2 - v_1 = 0.05\ mV$

$V_{out} = 5V$

$$\boxed{v_2 - v_1 = \frac{V_{out}}{A_v} = \frac{V_{out}}{10^6}}$$

Inverting amplifier,



$v_2 - v_1$ very small

$(v_2 \approx v_1)$ (few) (μV, mV) diff

$v_1 \approx v_2 = 0$, i from $v_1 = 0$, (Input impedance very high)

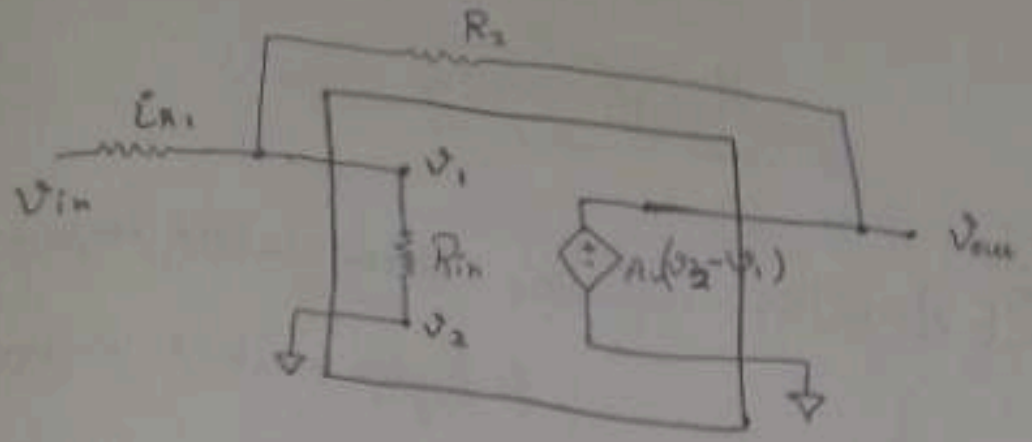
Virtual Ground ($v_1 \equiv$ not a real ground)

$$I_{R1} = \frac{v_{in} - 0}{R_1} = \frac{v_{in}}{R_1}, \quad I_{R2} = \frac{0 - V_{out}}{R_2} = -\frac{V_{out}}{R_2}$$

$$I_{R1} = I_{R2}$$

$$\Rightarrow \frac{v_{in}}{R_1} = \frac{-V_{out}}{R_2}$$

$$\Rightarrow \boxed{\frac{V_{out}}{v_{in}} = \frac{-R_2}{R_1}}$$



$$i_{R_1} = \frac{v_{in} - v_1}{R_1}, \quad i_{R_2} = \frac{v_1 - v_{out}}{R_2}$$

$$i_{R_1} = i_{R_2}$$

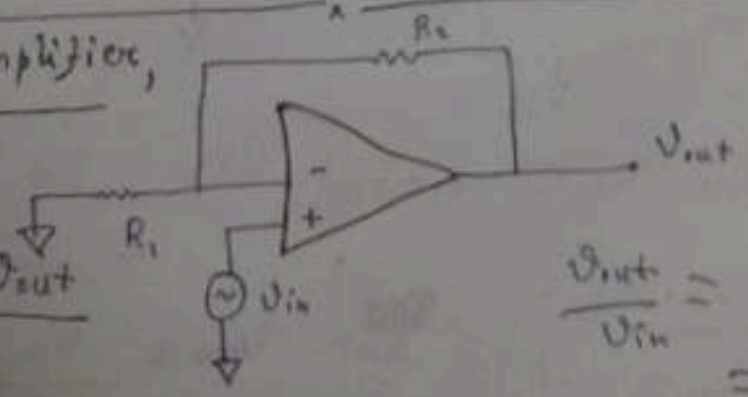
$$\Rightarrow \frac{v_{in} - v_1}{R_1} = \frac{v_1 - v_{out}}{R_2}$$

$$\Rightarrow v_{out} = -A_v v_1$$

$$\therefore \frac{v_{in} + \frac{v_{out}}{A_v}}{R_1} = \frac{-\frac{v_{out}}{A_v} - v_{out}}{R_2}$$

$$\frac{v_{out}}{v_{in}} = -\frac{R_2}{R_1 + \frac{R_2 + R_1}{A_v}} \approx \frac{-R_2}{R_1} \;//$$

Non-inverting Amplifier,

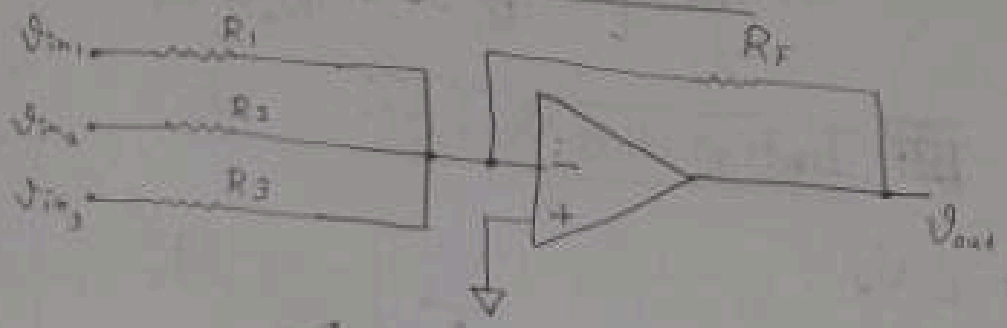


$$\frac{0 - v_{in}}{R_1} = \frac{v_{in} - v_{out}}{R_2}$$

$$\frac{v_{out}}{v_{in}} = R_2\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$= \left(1 + \frac{R_2}{R_1}\right) \;//$$

Summing Amplifier :

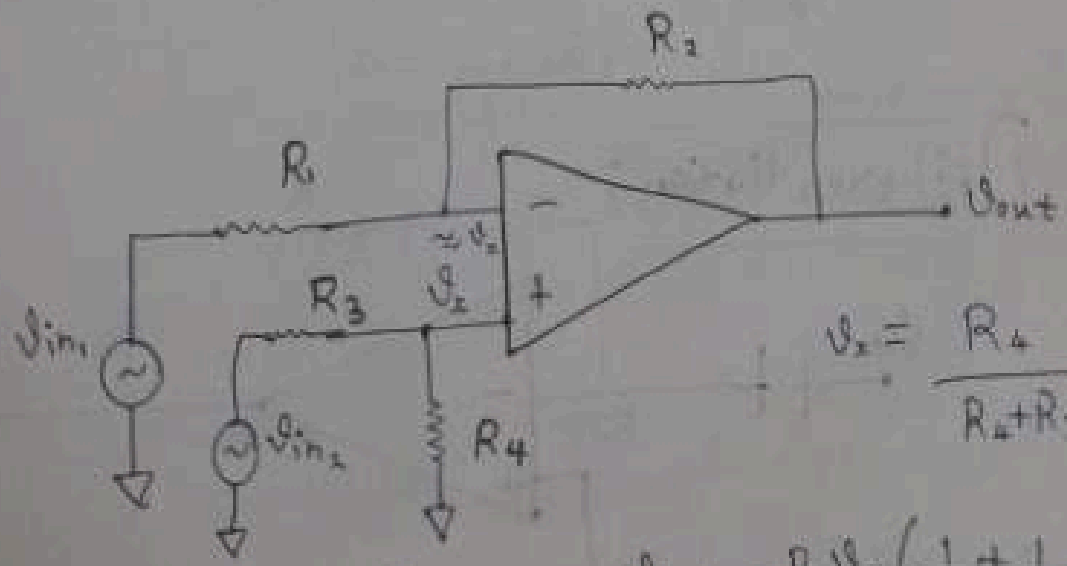


$$i_{R_F} = i_{R_1} + i_{R_2} + i_{R_3}$$

$$\frac{0 - v_{out}}{R_F} = \frac{v_{in_1} - 0}{R_1} + \frac{v_{in_2} - 0}{R_2} + \frac{v_{in_3} - 0}{R_3}$$

$$v_{out} = -\left(\frac{R_F}{R_1} v_{in_1} + \frac{R_F}{R_2} v_{in_2} + \frac{R_F}{R_3} v_{in_3}\right)$$

Difference Amplifier :



$$v_x = \frac{R_4}{R_4 + R_3} v_{in_2}$$

$$\frac{v_{in_1} - v_x}{R_1} = \frac{v_x - v_{out}}{R_2} \Rightarrow v_{out} = R_2 v_x \left(\frac{1}{R_1} + \frac{1}{R_2}\right) - \frac{v_{in_1} R_2}{R_1}$$

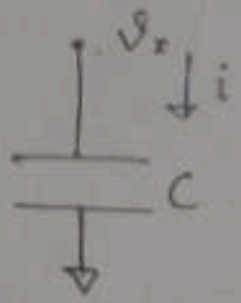
$$= \frac{R_2 R_4}{R_4 + R_3}\left(\frac{1}{R_1} + \frac{1}{R_2}\right) v_{in_2} - \frac{R_2}{R_1} v_{in}$$

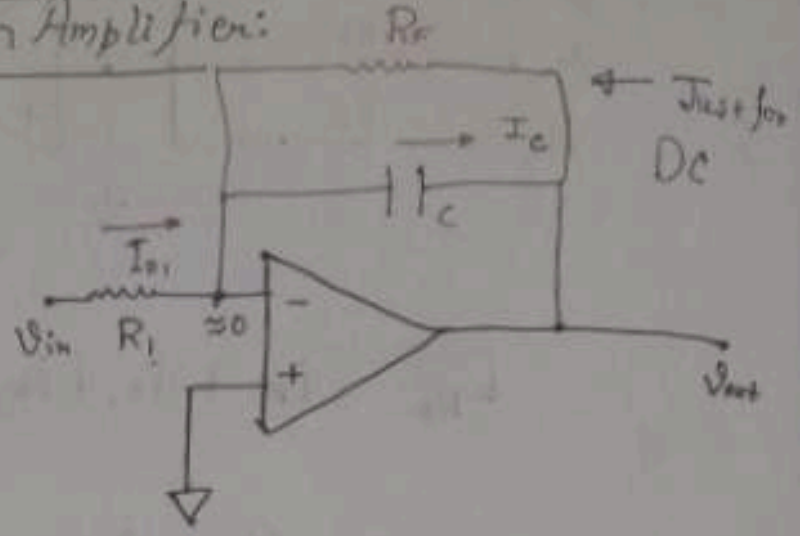
$$\Rightarrow v_{out} = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_4 + R_3}\right) v_{in_2} - \frac{R_2}{R_1} v_{in_1}$$

Integration Amplifier:

$$Q = C v_x$$

$$\frac{dQ}{dt} = C\frac{dv_x}{dt}$$





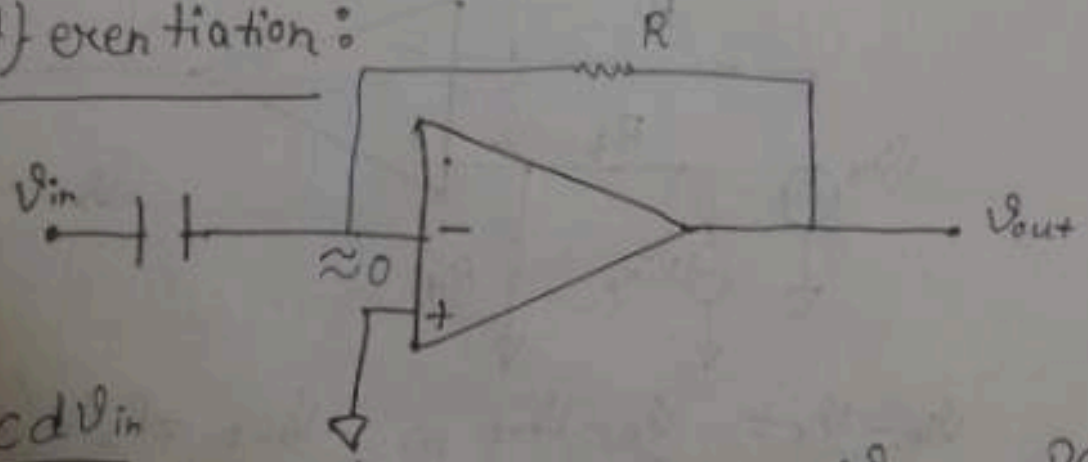
$$I_{R_1} = \frac{v_{in}}{R_1} = -C\frac{dv_{out}}{dt}$$

$$\int dv_{out} = -\frac{1}{R_1 C}\int v_{in}\, dt$$

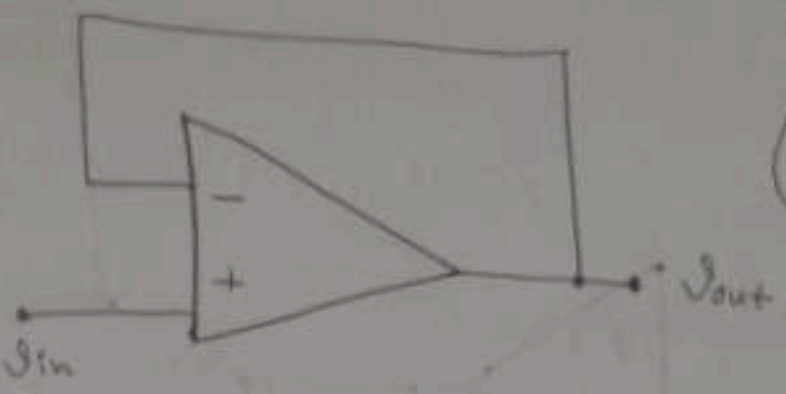
$$\boxed{v_{out} = v_{initial} - \frac{1}{CR_1}\int v_{in}\, dt}$$

$(v_{initial} = 0)$

Differentiation:



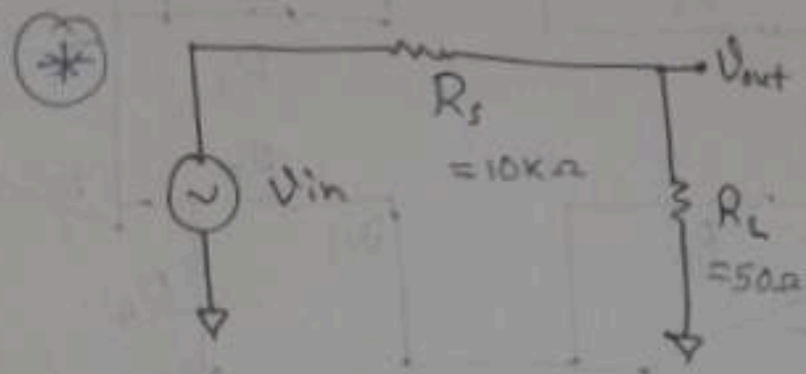
$$i = C\frac{dv_{in}}{dt} \Rightarrow -\frac{v_{out}}{R} = C\frac{dv_{in}}{dt} \Rightarrow v_{out} = -RC\frac{dv_{in}}{dt}$$



(Something to drive motors and all.)

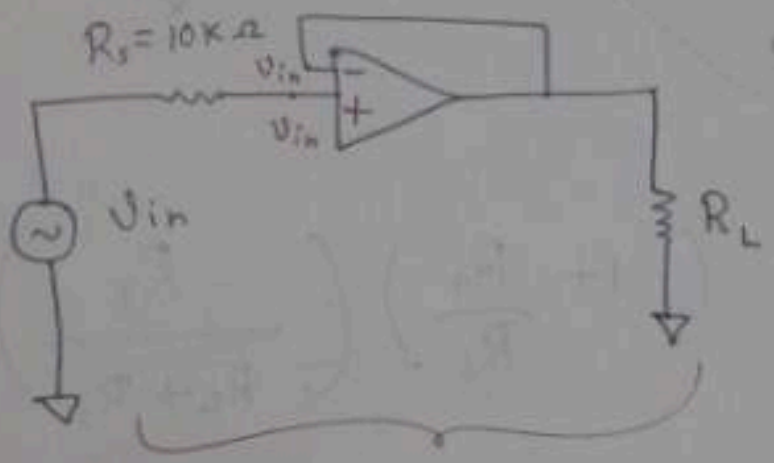
Voltage Follower:

$v_{in} = v_{out}$

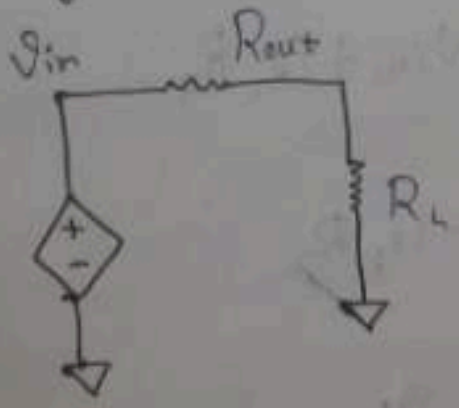


R_s very high

$$v_{out} = \frac{R_L v_{in}}{R_s + R_L}$$



(Voltage dependent voltage source)

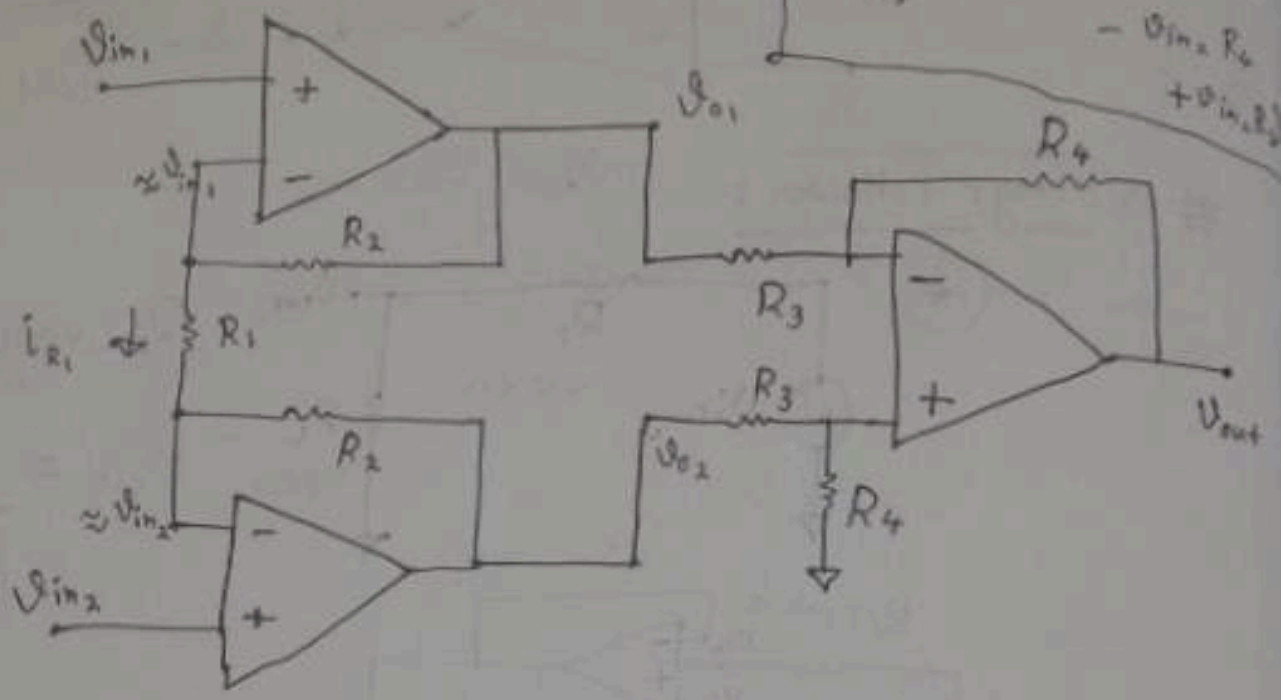


$$v_{out} = \frac{v_{in} R_L}{R_{out} + R_L}$$

$$\frac{R_4}{R_3}\left(v_{in_1} - \frac{v_{in_1}(R_3+R_4)}{R_3+R_4}\right.$$

$$\frac{R_4}{R_3}\left(R_3 v_{in_1} + R_4 v_{in_1} - v_{in_2} R_4 + v_{in_1} R_4\right)$$

#



$$v_{out} = \left(1 + \frac{R_4}{R_3}\right)\left(\frac{R_4}{R_4 + R_3}\right) v_{o2}$$

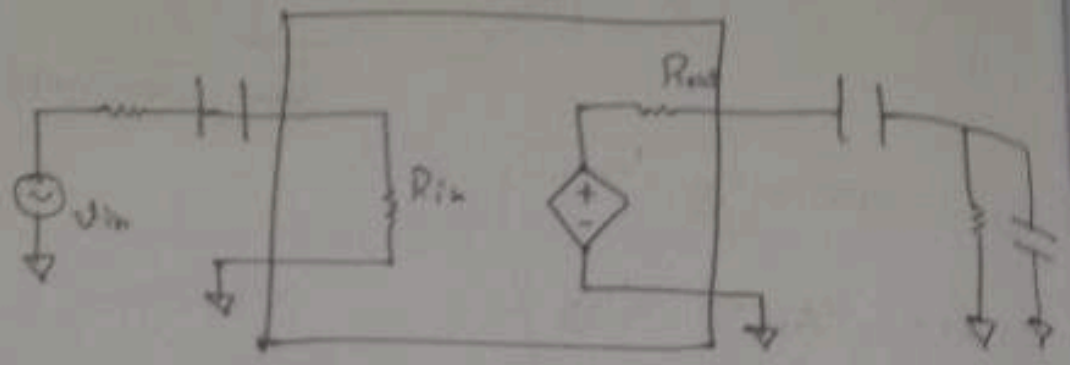
$$= \frac{R_4}{R_3}\left(v_{o2} - v_{o1}\right) - \frac{R_4}{R_3} v_{in_{o1}}$$

$$v_{o1} = \cancel{i_{R_1} R_1} \; v_{in_1} + i_{R_1} R_2$$

$$v_{o2} = v_{in_2} - i_{R_1} R_2 \qquad \left(i_{R_1} = \frac{v_{in_1} - v_{in_2}}{R_1}\right)$$

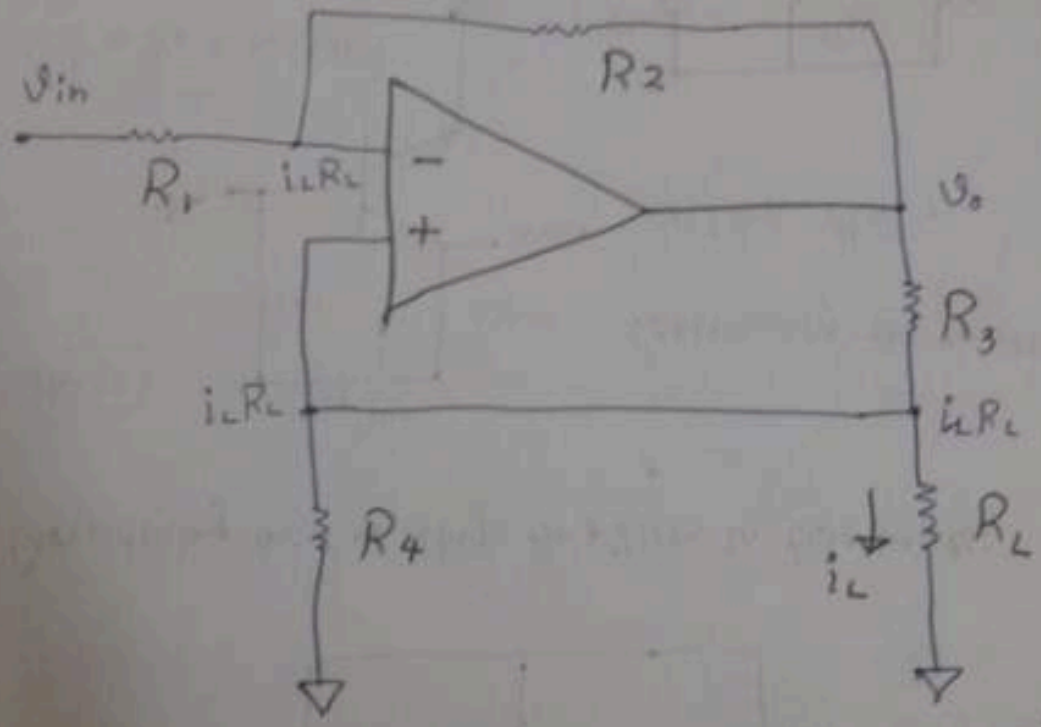
$$(v_{o1} - v_{o2}) = v_{in_1} - v_{in_2} + 2 i_{R_1} R_2$$

$$= (v_{in_1} - v_{in_2}) + 2\frac{(v_{in_1} - v_{in_2})}{R_1} R_2$$

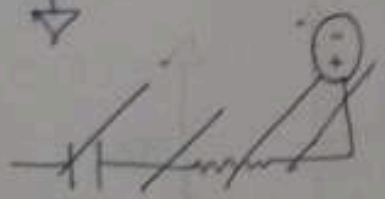


$$A_v' = A_v \frac{R_L}{R_L + R_{out}} \times \frac{R_{in}}{R_{in} + R_s}$$

Voltage to Current Converter:



Find the condition for which



i_L is independent of R_L.

$$\frac{v_{in} - i_L R_L}{R_1} = \frac{i_L R_L - v_o}{R_2} \rightarrow (1)$$

$$\frac{v_s - i_L R_L}{R_3} = \frac{i_L R_L}{R_4} + i_L \rightarrow (2)$$

$$-\left(\frac{v_{in} - i_L R_L}{R_1}\right)\frac{R_2}{R_3} = \frac{i_L R_L}{R_4} + i_L$$

$$\frac{R_2 v_{in}}{R_1 R_3} = \underbrace{\frac{i_L R_L R_2}{R_1 R_3}} - \underbrace{\frac{i_L R_L}{R_4}} - i_L$$

$$\cancel{\frac{i_L R_L R_2}{R_1 R_3}} = \cancel{\frac{i_L R_L}{R_4}}$$

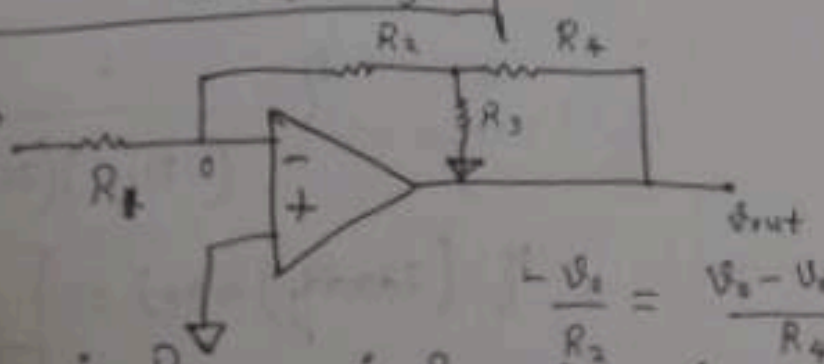
$$\Rightarrow \boxed{\frac{R_2}{R_1} = \frac{R_3}{R_4}} \qquad -i_L = \frac{v_{in} R_2}{R_1 R_3}$$

$$\boxed{i_L = \frac{-v_{in}}{R_4}}$$

$$\boxed{\begin{aligned} i_s &= \frac{v_s - i_L R_L}{R_2} - \frac{i_L R_L - V_o}{R_3} \\ &= (v_s - i_L R_L)\left(\frac{1}{R_2} + \frac{1}{R_3}\right) \end{aligned}}$$

$$i_{R_1} = \frac{v_{in}}{R_1}$$

\#



$$-\frac{v_s}{R_2} = \frac{v_s - v_{out}}{R_4} + \frac{v_o}{R_3}$$

$$v_s = -i_{R_2} R_2 \qquad \frac{i_{R_2} R_2}{R_2} = \frac{-i_{R_2} R_2 - v_{out}}{R_4} + \frac{-i_{R_2} R_2}{R_3}$$

For LED,

from graph 1; $m = \frac{dI_D}{dV_D} = \frac{(8.6 - 6.7)\,mA}{(2030 - 2010)\,mV} = 0.095\ A^{-1}$

$$r_{on} = \frac{dV_D}{dI_D} = \frac{1}{m} = \boxed{10.526\ \Omega}$$

from graph 2;

$$m' = \frac{(-1.203) - (0.6418)}{(1790 - 1770)\,mV} = \frac{0.0214}{10^{-3}} = \boxed{21.5}$$

∴ using $y = m'x + c$ for,

$$\ln(I_D) = \frac{V_D}{\eta V_T} + \ln I_S$$

$$c = \ln I_S = \ln(8.6 \times 10^{-3}) - 21.5 \times 2030 \times 10^{-3}$$

$$= -48.4$$

$$I_S = \boxed{9.55 \times 10^{-22}\ A}$$

$$\text{Slope} = \frac{1}{\eta V_T} = 21.5$$

$$\eta = \frac{1}{(21.5)(26 \times 10^{-3})}$$

$$\eta\ (\text{Identity factor}) = \boxed{1.788}$$

$$\frac{v_{out}}{R_4} = -\left(1 + \frac{R_2}{R_4} + \frac{R_2}{R_3}\right) i_{R_2}$$

$$v_{out} = -\frac{R_4}{R_1}\left(1 + \frac{R_2}{R_4} + \frac{R_2}{R_3}\right) v_{in}$$

$$= -\left(\frac{R_4}{R_1} + \frac{R_2}{R_1} + \frac{R_2 R_4}{R_1 R_3}\right) v_{in}$$

For Silicon Diode,

from graph 1 ⇒ $m = \frac{dI_D}{dV_D} = \frac{(16.05 - 7.43)\,mA}{(685 - 671)\,mV} = 0.6293\ \Omega^{-1}$

Forward resistance $r_{on} = \frac{dV_D}{dI_D} \approx \frac{1}{m} = \boxed{5.476\ \Omega}$

from graph 2 ⇒ $m' = \frac{(-2.38) - (-4.49)}{(453 - 261)\,m} = 0.0228$

$= 22.6$

∴ using $y = m'x + c$, for

$$\ln(I_D) = \frac{V_D}{\eta V_T} + \ln I_S$$

$$\ln(I_S) = -\frac{(22.6)}{22.6} \times 685 \times 10^{-3} + \ln(1.05 \times 10^{-3})$$

$$I_S = \boxed{1.9 \times 10^{-9}\ A} = -20.08$$

Slope → $m' = \frac{1}{\eta V_T} = 22.6$

∴ $\eta = \frac{1}{m' V_T} = \frac{1}{(22.6)(26 \times 10^{-3})}$

$(V_T = 26\,mV)$

(Identity factor) $\boxed{\eta = 1.70}$